

Random Cluster Model on Regular Graphs

Limit of pressure function for a sequence of random regular graphs

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Introduction

- In statistical physics models, such as Ising, Potts, random cluster, it is often insightful to look at the “Pressure function”.
- In probability theory, we study these models on a wide class of graphs.
- The random cluster model is a generalization of bond percolation with dependent edges/bonds.
- Often on a graph sequence, we study the properties of the pressure function (such as differentiability) to infer about the model in general.
- Here, we look at a sequence of random regular graphs and the partition function and pressure function of the random cluster model on them.
- Has applications not only in physics but also in synthetic and real-world network data analysis.

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Random Cluster Model

Definition (Random Cluster)

- For a finite graph $G = (V, E)$, we assign to a specific edge-subset A , of open edges, the probability proportional to $q^{k(A)} w^{|A|}$, where, $k(A)$ = the number of connected components of the graph (V, A) .
- The partition function of the random cluster model is defined to be the normalizing constant, i.e.,

$$Z_G(q, w) = \sum_{A \subseteq E} q^{k(A)} w^{|A|} .$$

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Main Result

Theorem (Bencs, Borbényi, Csikvári, 2023)

If $(G_n)_n$ is a sequence of random d -regular graphs, then the limit exists and is identified as follows:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln Z_{G_n}(q, w) = \ln \Phi_{d,q,w}$$

with probability 1, for any $q \geq 2$ and $w \geq 0$.

$$\begin{aligned} \Phi_{d,q,w} = \max_{t \in [-\pi, \pi]} & \left[\left(\sqrt{1 + \frac{w}{q}} \cos(t) + \sqrt{\frac{(q-1)w}{q}} \sin(t) \right)^d \right. \\ & \left. + (q-1) \left(\sqrt{1 + \frac{w}{q}} \cos(t) - \sqrt{\frac{w}{q(q-1)}} \sin(t) \right)^d \right] \end{aligned}$$

Large Girth Graphs

Definition (Essentially large girth)

Call a sequence of graphs (G_n) with $|V(G_n)| \rightarrow \infty$, essentially large girth sequence of graphs if

$$\lim_{n \rightarrow \infty} \frac{L(G_n, g)}{|V(G_n)|} = 0 \text{ for all positive integers } g,$$

where $L(G, g)$ denotes the number of cycles in G of the size at most $(g - 1)$.

Main result

- It is known that a sequence of random d -regular graphs is essentially large-girth graph sequence with probability 1. Taking one vertex at random, if we look at a local neighbourhood around it, it looks like a tree for most of the vertices.
- It can be shown that for d -regular large girth graphs, the limiting quantity $\Phi_{d,q,w}$ enjoys similar properties like discontinuity, non-differentiability at specific values of the parameters.
- The trigonometric objective function comes from "Belief propagation".

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Bethe Entropy and Bethe Functional

- In statistical physics, belief propagation is a recursive algorithm to calculate the partition function. It can be shown that the fixed-point solutions, in case of tree graphs, give the original partition function.
- It is hence believed that in case of locally tree-like graphs too, the fixed points correspond to a sensible (under-) estimate, for the partition function, called Bethe free energy.
- The objective function comes from a functional by Bethe, calculated for d -regular graphs, which by some suitable transformation gives the trigonometric objective function and the set of fixed points gets reparameterized to the interval $[-\pi, \pi]$.

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r - Spin Model

Definition (r - Spin Model)

We call a model a spin model with r -spins if there is an $r \times r$ symmetric matrix N , and a vector $\underline{\mu} \in \mathbb{R}^r$ such that for a graph $G = (V, E)$ the probability assigned to a $\sigma : V \rightarrow \{1, 2, \dots, r\}$ is

$$P(\sigma) = \frac{1}{Z_G(N, \underline{\mu})} \prod_{v \in V} \mu_{\sigma(v)} \prod_{(u,v) \in E} N_{\sigma(u), \sigma(v)} ,$$

and the partition function is defined as

$$Z_G(N, \underline{\mu}) = \sum_{\sigma: V \rightarrow [r]} \prod_{v \in V} \mu_{\sigma(v)} \prod_{(u,v) \in E} N_{\sigma(u), \sigma(v)} .$$

- **Remark.** For integer q , coupling between the random cluster model with parameters q and w , and the q - spin model with $N = wI_q + \mathbb{1}_q \mathbb{1}_q^T$, $\underline{\mu} = \mathbb{1}_q$. In general when the $\underline{\mu}$ will be the vector with all entries 1 then, $Z_G(N, \underline{\mu})$ denoted simply with $Z_G(N)$.

Approximating the Matrix N : Rank 1 approximation

- Consider the random cluster model with $q \in \mathbb{N}$. Take $N = wI_q + \mathbb{1}_q \mathbb{1}_q^T$, by coupling, $Z_G(N) = Z_G(q, w)$.
- Approximate N by the rank 1 matrix, $M_1 = (1 + \frac{w}{q}) \mathbb{1}_q \mathbb{1}_q^T$.
- Define 'Rank 1 approximation' : $Z_G^{(1)}(q, w) := Z_G(M_1)$.
- If $q \geq 1$, Then, rank 1 approximation is an under-approximation

$$Z_G^{(1)}(q, w) \leq Z_G(q, w) \quad .$$

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Approximating the Matrix N : Rank 2 approximation

- Approx. N by the rank 2 matrix

$$M_2 = \begin{pmatrix} 1+w & 1 & \dots & 1 \\ 1 & 1+\frac{w}{q-1} & \dots & 1+\frac{w}{q-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1+\frac{w}{q-1} & \dots & 1+\frac{w}{q-1} \end{pmatrix}$$

- Define 'Rank 2 approximation' : $Z_G^{(2)}(q, w) := Z_G(M_2)$.

Approximating the Matrix N : What if q non-integer ?

- Take

$$M'_2 = \begin{pmatrix} 1+w & 1 \\ 1 & 1+\frac{w}{q-1} \end{pmatrix} \quad \underline{\mu}'_2 = \begin{pmatrix} 1 \\ q-1 \end{pmatrix} \quad .$$

Then, by simple calculation,

$$Z_G(M_2) = Z_G(M'_2, \underline{\mu}'_2) \quad .$$

- This can be extended even for non-integer q 's.

$$\boxed{Z_G^{(2)}(q, w) := Z_G(M'_2, \underline{\mu}'_2)}$$

- Random cluster model approximated by a 2-spin model.

How good is rank 2 approximation?

Theorem (Bounds for rank 2 approximation)

For $q \geq 2$,

$$Z_G^{(2)}(N, \mu) \leq Z_G(N, \mu) \leq q^{n/g+L} Z_G^{(2)}(N, \mu)$$

for all $g \in \mathbb{N}$. Recall, L is the number of cycles of size at most $(g-1)$.

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A theorem for the sequence of pressure functions

Theorem (Pressure function sequence convergence)

Let $q \geq 2$ and $w \geq 0$. Let $(G_n)_n$ be a sequence of random d -regular graphs. If the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln Z_{G_n}^{(2)}(q, w),$$

exists (in a.s. sense), then the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln Z_{G_n}(q, w),$$

exists too (in a.s. sense), and they have the same value.

Proof.

$$\begin{aligned} Z_{G_n}^{(2)}(N, \mu) &\leq Z_{G_n}(N, \mu) \leq q^{n/g+L} Z_{G_n}^{(2)}(N, \mu) \\ \implies \frac{1}{n} \ln Z_{G_n}^{(2)}(N, \mu) &\leq \frac{1}{n} \ln Z_{G_n}(N, \mu) \leq \left(\frac{1}{g} + \frac{L}{n} \right) \ln q + \frac{1}{n} \ln Z_{G_n}^{(2)}(N, \mu) \end{aligned}$$

As the sequence is essentially large girth, as $\frac{L}{n}$ goes to 0, and this is true for every positive integer g , and as the limit exists, so conclude. $\square$

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The matrix M'_2

- Recall, $M'_2 = \begin{pmatrix} 1+w & 1 \\ 1 & 1+\frac{w}{q-1} \end{pmatrix}$
- As rank 2, pd, can be written in the form $\underline{a}\underline{a}^T + \underline{b}\underline{b}^T$
- Take

$$\underline{a} = \begin{pmatrix} \sqrt{1 + \frac{w}{q}} \\ \sqrt{1 + \frac{w}{q}} \end{pmatrix} \quad \text{and} \quad \underline{b} = \begin{pmatrix} \sqrt{\frac{(q-1)w}{q}} \\ -\sqrt{\frac{w}{q(q-1)}} \end{pmatrix}.$$

- Note that If (a, b) are a valid pair of decomposition for M'_2 , Then the following is also valid,

$$a_j(t) = a_j \cos(t) + b_j \sin(t)$$

$$b_j(t) = -a_j \sin(t) + b_j \cos(t)$$

where $j \in \{1, 2\}$

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Belief Propagation Revisited

- In case of an $(N, \underline{\mu})$ spin model on random d -regular graphs, which have locally tree like structure, the Bethe belief propagation recursion is given by: $BP(h)_\sigma = \mu_\sigma \left(\sum_{\sigma'} N_{\sigma, \sigma'} h(\sigma') \right)^{d-1}$, where, h is a probability distributions in $\mathbb{R}^r$.
- The Bethe functional is defined as :

$$\tilde{\Phi}_{d, N, \mu}(h) = \log \left[\sum_{\sigma} \mu_{\sigma} \left(\sum_{\sigma'} N_{\sigma, \sigma'} h_{\sigma'} \right)^d \right] - \frac{d}{2} \log \left[\sum_{\sigma, \sigma'} N_{\sigma, \sigma'} h_{\sigma} h_{\sigma'} \right] .$$

- The Bethe free energy, which serves as the estimate of the partition function is defined as:

$$\Phi_d(N, \mu) = \sup_{h \in \mathcal{H}} \exp(\tilde{\Phi}_{d, N, \mu}(h)) .$$

Where, $\mathcal{H}$ is the set of fixed points of the Bethe belief propagation recursion.

- Recall,

$$\tilde{\Phi}_{d,N,\underline{\mu}}(h) = \log \left[\sum_{\sigma} \mu_{\sigma} \left(\sum_{\sigma'} N_{\sigma,\sigma'} h_{\sigma'} \right)^d \right] - \frac{d}{2} \log \left[\sum_{\sigma,\sigma'} N_{\sigma,\sigma'} h_{\sigma} h_{\sigma'} \right] .$$

- Considering a 2×2 matrix and a vector in $\mathbb{R}^2$, substituting $R = \frac{h_1}{h_2}$,

$$\begin{aligned} & \exp \left(\tilde{\Phi}_{N,\underline{\mu},d}(h) \right) \\ &= \mu_1 \left[\frac{N_{11}R + N_{12}}{\sqrt{N_{11}R^2 + 2N_{12}R + N_{22}}} \right]^d + \mu_2 \left[\frac{N_{12}R + N_{22}}{\sqrt{N_{11}R^2 + 2N_{12}R + N_{22}}} \right]^d . \end{aligned}$$

- Putting values $N = M'_2$, $\mu = \mu'_2$, and by the $(a(t), b(t))$ decomposition of M'_2 , the quantity reduces to our desired trigonometric objective function and $\mathcal{H}$ corresponds to the interval $[-\pi, \pi]$. We represent the $\Phi_d(N, \underline{\mu})$ term as $\Phi_{d,q,w}$ in this context.

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Evaluation of the limit

Theorem (Dembo, Montanari, Sly, Sun, 2014)

Let G be a random d -regular graph on n vertices. Then,

$$EZ_G(N, \mu) = n^{O(1)} \Phi_d(N, \mu)^n \quad .$$

Evaluation of the limit

Theorem (Ruozzi, 2012)

For a 2×2 pd matrix N with positive entries and $\mu \in \mathbb{R}_{\geq 0}^2$ and a d -regular graph G we have $Z_G(N, \mu) \geq \Phi_d(N, \mu)^{|V(G)|}$.

Evaluation of the limit

Theorem (Bencs, Borbényi, Csikvári, 2023)

For any large girth sequence of d -regular graphs $(G_n)_n$ the sequence of pressure functions, i.e.

$$\frac{1}{n} \ln Z_{G_n}(N, \underline{\mu})$$

converges.

Evaluation of the limit

Theorem (Bencs, Borbényi, Csikvári, 2023)

Let N be a 2×2 pd matrix with positive entries and $\underline{\mu} \in \mathbb{R}_{>0}^2$. if $(G_n)_n$ is a sequence of random d -regular graphs, then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln Z_{G_n}(N, \underline{\mu}) = \ln \Phi_d(N, \underline{\mu})$$

with probability 1.

Proof.

By Dembo, $\mathbb{E}Z_{G_n}(N, \underline{\mu}) = n^C \Phi_d(N, \underline{\mu})^n$ again, by Ruozzi, $Z_{G_n}(N, \underline{\mu}) \geq \Phi_d(N, \underline{\mu})^n$.

$$\begin{aligned} & \left| \frac{1}{n} \ln Z_{G_n}(N, \underline{\mu}) - \ln \Phi_d(N, \underline{\mu}) \right| > \varepsilon > 0 \\ \iff & \frac{1}{n} \ln Z_{G_n}(N, \underline{\mu}) - \left[C \frac{\ln n}{n} + \frac{1}{n} \ln \mathbb{E}Z_{G_n}(N, \underline{\mu}) \right] > \varepsilon \\ \iff & \frac{1}{n} \ln \frac{Z_{G_n}(N, \underline{\mu})}{\mathbb{E}Z_{G_n}(N, \underline{\mu})} > \varepsilon + C \frac{\ln n}{n} \\ \iff & \ln \frac{Z_{G_n}(N, \underline{\mu})}{\mathbb{E}Z_{G_n}(N, \underline{\mu})} > n\varepsilon - C \ln n \\ \iff & \frac{Z_{G_n}(N, \underline{\mu})}{\mathbb{E}Z_{G_n}(N, \underline{\mu})} > \frac{e^{n\varepsilon}}{n^C} \quad \dots (*) \end{aligned}$$

The probability of (*) by Markov is at most $\frac{n^C}{e^{n\varepsilon}}$ which is summable by ratio test as $\varepsilon > 0$., Hence, by Borel - Cantelli, we conclude.



Thank You!

Any Questions?